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Design and Simulation of 1W Audio Amplifier

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ABSTRACT Power amplifiers are vital electronics devices, responsible for amplifying small signals to an amplified level that is usable by other devices. This power amplifier design is intended as a consumer audio device and must abide by pre-determined constraints and design parameters for marketing incentives. Such parameters relate to the sound quality, power output, size, power efficiency, and compatibility of the audio amplifier. The primary component that the design used is the LM386 low voltage power amplifier. The audio amplifier design focused on meeting the design parameters by applying a sensitivity analysis to passive components. The design utilized simulation software, LTspice and MATLAB, to assist in the design process. Final component values are chosen based on sensitivity analysis and are simulated with the final values. The final performance characteristics of the circuit are measured and compared to the design requirements. The final design is crafted with an electronic computer-aided design (ECAD) tool so that the product is ready to be produced as a printed circuit board (PCB).

INDEX TERMS Audio amplifier, product design, sensitivity analysis

I. INTRODUCTION

From [1], the design is based on the "LM386 with Gain = 20" topology from the LM386 datasheet [2]. The circuitry and passive component values may deviate from [2] due to the design parameter requirements [1].

A constraint of the audio amplifier is that it must use a single LM386 amplifier. The LM386 amplifier from Texas Instruments is a class-AB audio amplifier specified for low voltage consumer applications [2]. With this constraint, the LM386 is modeled for LTspice simulations, and the LM386M-1 model is used by an electronic computer-aided design (ECAD) tool.

Table I from [1] states specific design parameters that the audio amplifier and the justification for the correlating parameter. Total harmonic distortion (THD) is a waveform related measurement, where sound quality and volume are affected by this parameter. Sound quality of the amplifier is also affected by noise, measured as input-referred noise (IRN) [1]. Power supply rejection ratio (PSRR) and common mode rejection ratio (CMRR) are rejection ratio measurements, where the sound quality is affected by both parameters. The gain of the amplifier should be consistent over hearable frequencies for good sound quality. The maximum printed circuit board (PCB) size is 1.25 by 2.0 inches, to keep the amplifier small and cheap. The efficiency of the amplifier at 1% THD should be greater than 40%, which is common with similar amplifiers. The maximum voltage gain should be greater than 26dB to be compatible with connected devices. The alternating current (AC) input and output impedances of the amplifier should be sized correctly to be compatible with connected devices. Power efficiency is also affected by the AC output impedance. The amplifier should function with any supply voltage between 6-12V because they are typical input supply voltages for this type of audio amplifier.

TABLE I
AUDIO AMPLIFIER DESIGN SPECIFICATIONS [1]

Parameter	Justification
total harmonic distortion (THD) < 1% up to the 5th harmonic [a]	the human ear can detect distortion greater than 1% THD and 1W can produce an adequate volume
input-referred noise < 75nV/√Hz from 100Hz to 20kHz	the human ear can easily detect noise if the signal to noise ratio is less than 40dB
power supply rejection ratio > 60 dB from 100Hz to 20kHz	
common mode rejection ratio > 60 dB from 100Hz to 20kHz	
gain is within ± 3dB from 100Hz to 20kHz [b]	this will provide consistency over the range of human hearing
max circuit board size is 1.25 x 2.0 inches	will fit within a typical portable speaker enclosure
efficiency > 40% at P_{max} power output (the power at which THD=1%) [c]	other amplifiers of this class can achieve this. It is significantly lower than the theoretical maximum efficiency.
max voltage gain ≥ 26 dB	most portable devices have signal levels above 500mV
AC input impedance > 1kΩ	compatibility with headphone or line-out sources
AC output impedance < 1Ω	for compatibility with speakers
functions with supply voltages between 6V and 12V	typical DC power supply voltages

[a]Assumes that it uses a 1kHz input, 8Ω load, and is at 1W output power.
[b]Assumes an 8Ω load.
[c]Assumes a 1kHz input, 8Ω load, and 12V supply.

The ideal simulation software for measuring these design parameters is a Simulation Program with Integrated Circuit Emphasis (SPICE) tool, specifically LTspice. This is because LTspice can model complicated integrated circuits (ICs) and has the capability to run different types of simulations. To simulate each design parameter from Table I, two circuits are required.

Fig. 1 is an LTspice schematic of the modeled audio amplifier system [3]. This schematic can measure most of the design parameters for the sensitivity analysis.

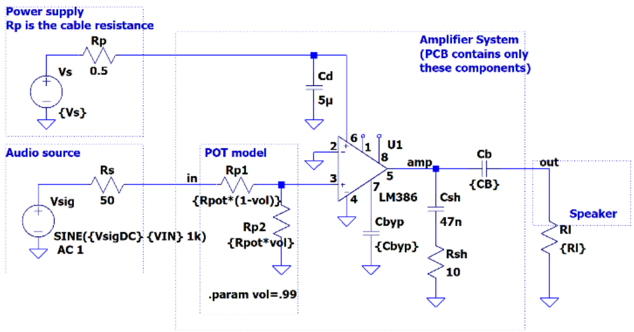

FIGURE 1. **LTspice schematic for sensitivity analysis [3]. Used to measure cutoff frequencies, input-referred noise, and maximum power out.**

Fig. 2 shows an LTspice schematic that measures the rejection ratio parameters [4]. An individual schematic for rejection ratio is required because it is the ratio of two different gains. The unwanted gain can only be measured by an altered circuit. This paragraph is refined by [5].

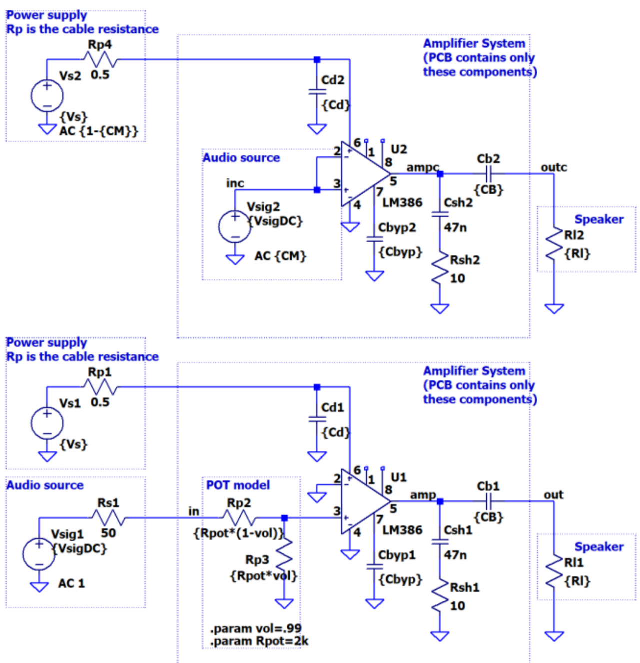

FIGURE 2. **LTspice schematic for sensitivity analysis [4]. Used to measure common mode rejection ratio and power supply rejection ratio.**

II. SIMULATION SETUP

A. SYSTEM MODELING

The four sections of the audio amplifier system model are the power supply, audio source, amplifier system, and speaker [3].

The amplifier system is based on the schematic shown in [2], with an added bypass capacitor. The LM386 IC chip is modeled into the circuit with the correlating datasheet values. The potentiometer is modeled by a voltage divider, where the resistance values are inverse of each other.

The power supply is modeled as a voltage source and resistor for the cable's resistance. The audio source is modeled as a voltage source and a 50Ω resistor modeling the source resistance. The speaker impedance is frequency dependent, but it is modeled as a resistor for the simulations [3].

B. COMPONENTS

The components that are evaluated during the design process are C_{byp}, C_b, V_S, $V_{sig\text{-}DC}$, R_{load}, and R_{pot}.

C_{byp} is the bypass capacitor placed on pin seven of the LM386, the bypass decoupling path [2]. The value of bypass capacitors can affect performance characteristics of the amplifier [3]. C_b is the blocking capacitor, used to attenuate the direct current (DC) portion of the signal before reaching the load. V_s is the power supply voltage, responsible for powering the audio amplifier. This voltage input goes to pin six of the LM386 [2]. $V_{sig\text{-}DC}$ is the DC voltage for the input signal to vary around. R_{load} is the impedance of the speaker that is hooked up to the audio amplifier. R_{pot} is the max resistance value that the potentiometer possesses. The potentiometer is located between the input and pin three of the LM386.

C. SIMULATED PARAMETERS

There are six parameters that are tested during the sensitivity analysis. These are high pass cutoff frequency, f_{HP}, low pass cutoff frequency, f_{LP}, CMRR, PSRR, IRN, and maximum output power (MOP) at 1% THD.

Cutoff frequency is a location of a filter where there is a change in the gains slope. The high pass cutoff frequency, f_{HP}, is the frequency of the amplifier where lower frequencies are attenuated and higher frequencies are passed. The low pass cutoff frequency, f_{LP}, is the frequency of the amplifier where higher frequencies are attenuated and lower frequencies are passed.

Rejection ratio is the ability to amplify desired signals against unwanted signals. CMRR is ratio of regular gain over the common mode gain [4]. Common mode gain is from when there is no power supply, and the input terminals of the amplifier are shorted [4]. Power supply gain is gain from when there is no input signal and only voltage from the power supply [4]. Both gains are considered unwanted.

Circuit noise is a property of any electronic that can use power [6]. IRN is the measurement of how much an input signal is affected by a circuit's noise [7].

MOP is the highest power output before THD is above 1% for a 1kHz signal. THD measures the deformity of a signal by comparing the unwanted harmonics of the waveform to the intended harmonic. From [1], THD up to the first five harmonics is calculated by

$$\%THD = \frac{\sqrt{(V_2^2 + V_3^2 + V_4^2 + V_5^2)}}{V_1^2} * 100 \qquad (1)$$

D. MODELING FUNCTIONS

The AC simulation takes measurements of gain and phase with respect to frequency. This simulation is helpful for finding f_{HP}, f_{LP}, CMRR, and PSRR. Transient simulations measure a signal with respect to time, responsible for MOP calculations. The noise simulation is a specialized simulation that measures noise with respect to frequency. This simulation is helpful for measuring the audio amplifiers IRN.

The LTspice simulations utilized measure, step, and Fast Fourier Transform (FFT) statements. The measure statement helped log data and assign it to a variable. Step statements are used for simulating multiple values of a given component in a single simulation. The Fourier statement is used for taking the FFT of a transient waveform and outputting the voltage values for each harmonic [8]. This is specifically helpful for the MOP parameter, ensuring it is measured at 1% THD.

III. SIMULATION RESULTS

This section will interpret data collected from the sensitivity analysis, examine significant graphs, discuss component value selection, and cover PCB design.

A. SENSITIVITY ANALYSIS

Table II is an organizer that provides information on how a component affects a design parameter.

TABLE II
SENSITIVITY TABLE

Parameter	C_{byp}	C_b	V_S	$V_{sig\text{-}DC}$	R_{load}	R_{pot}
f_{HP}	~	↓	~	~	↓	X
f_{LP}	~	~	↑	↑	↑	X
CMRR	↑	~	↑	↑	~	X
PSRR	↑	~	~	~	~	X
IRN	~	~	↓	↓	~	↓↑
MOP	~	↑	↑	↑↓	↓	X

The symbols used in Table II are ~, ↑, ↓, and X. The neutral symbol, ~, indicates that a change in the component value does not significantly affect the parameter. The direct relationship symbol, ↑, indicates that an increase in the component value increases the parameter value. The inverse relationship symbol, ↓, indicates that an increase in the component value decreases the parameter value. The untested symbol, X, means that the component value was not tested in comparison with the parameter. A cell that has both a direct and inverse relationship symbol means that a change in the components' value can either increase or decrease the parameter.

1) CAPACITANCES

The bypass capacitor, C_{byp}, only can affect the rejection ratio parameters. Since C_{byp} has a direct relationship with CMRR and PSRR, a large C_{byp} will increase the amplifiers' efficiency at amplifying the desired signal from unwanted signals.

Blocking capacitor, C_b, has an inverse relationship with high pass cutoff frequency and a direct relationship with MOP.

2) VOLTAGES

Power supply voltage, V_S, has direct relationships with the low pass cutoff frequency, CMRR, and MOP, and it has an inverse relationship with IRN. This means a higher V_S will increase CMRR and decrease IRN.

Input signal DC voltage, $V_{sig\text{-}DC}$, has direct relationships with the low pass cutoff frequency and CMRR, and it has an inverse relationship with IRN. Notably, $V_{sig\text{-}DC}$ has a direct relationship with MOP from values -0.2 to -0.04, and an inverse relationship with MOP from values -0.04 to 0.2.

3) RESISTANCES

Load resistance, R_L, has a direct relationship with the low pass cutoff frequency, and inverse relationships with the high pass cutoff frequency and MOP.

Potentiometer resistor, R_{pot}, has a mixed relationship with IRN and plotted in Fig 3. R_{pot} has an inverse relationship with IRN from 2kΩ to around 12.5kΩ and a direction relationship from around 12.5kΩ to 50kΩ.

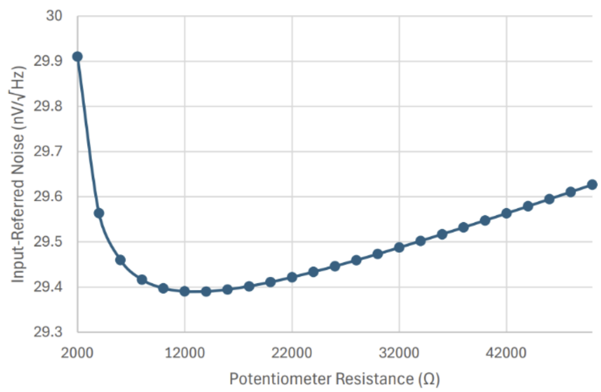

FIGURE 3. Plotted relationship between potentiometer resistance (Ω) and input-referred noise (nV/√Hz).

B. COMPONENT SELECTION

Table III is a table of the component values that will be used in the final design.

TABLE III
FINAL COMPONENT VALUES

Component	C_{byp} (F)	C_b (F)	V_S (V)	$V_{sig\text{-}DC}$ (V)	R_{load} (Ω)	R_{pot} (Ω)
Final Value	10u	220u	12	0	8	12.5k

V_S and R_{load} were chosen based on previous assumptions made from Table I. R_{pot} is 12.5kΩ because IRN is only affected by it, and it is resistance value with the lowest IRN. An increase in C_{byp} will only benefit the amplifier due to an increase in CMRR and PSRR, however is constrained by 10µF from [1], so 10µF is chosen. Similarly, an increase in C_b only will benefit the amplifier because the high pass cutoff frequency will decrease and MOP will increase, which will ensure sound quality at low frequencies and a higher output power respectively. $V_{sig\text{-}DC}$ has benefits and disadvantages, so 0V is chosen for simplicity.

Fig. 4 is a plot of P_{out} vs. THD using the final component values. An increase in power to the load increases the distortion of the waveform, especially at the 1W mark.

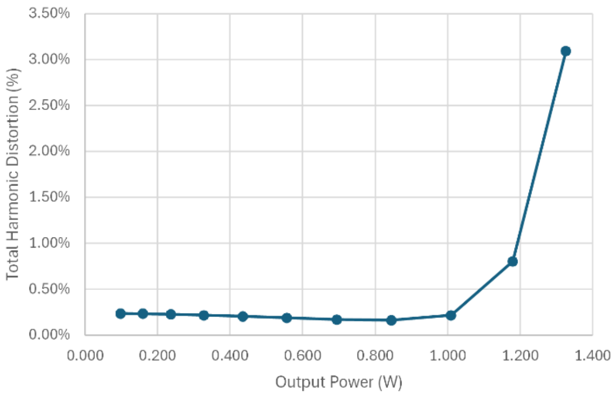

FIGURE 4. Plot of Pout vs THD using final component values, simulated with LTspice.

C. PRINTED CIRCUIT BOARD DESIGN

The audio amplifier design was translated to an Electronic Computer-Aided Design (ECAD) tool to have the product PCB ready. Fig. 5 is the PCB layout for the audio amplifier, drafted by Altium Designer, with the help of [9].

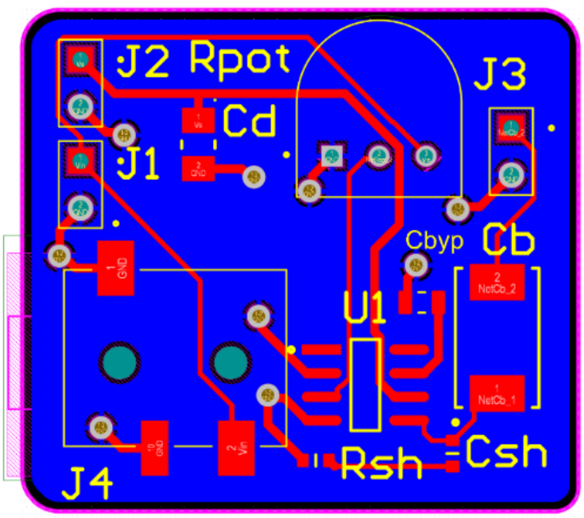

FIGURE 5. Printed circuit board (PCB) layout of audio amplifier using Altium Designer.

IV. SIMULATION AND CONCLUSION

Fig. 5 is an LTspice schematic of the audio amplifier with the final component values from Table III. This schematic is responsible for all measurements in Table IV, with additional schematic requires for the rejection ratios.

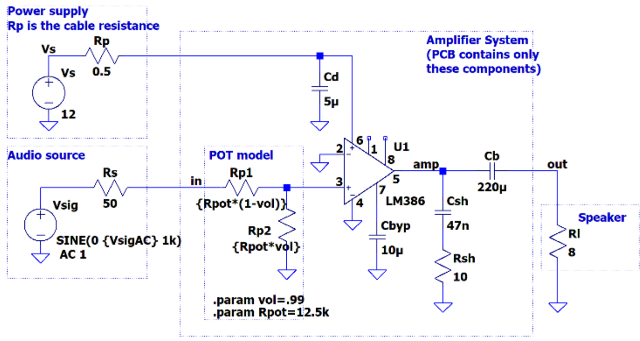

FIGURE 5. LTspice schematic with the final component values.

Table IV compares the design parameters and the measured parameter values for the audio amplifier. The measured THD at 1W, with an 8Ω load and 1kHz input is 0.21%, below the 1% threshold. IRN measured from 100Hz to 20kHz is 26.8nV/√Hz, below the 75nV/√Hz threshold. PSRR and CMRR are 65.6dB and 62.6dB respectively, greater than the 60dB threshold. Gain consistency is measured as a range, obtained from when the gain is 3dB below the max gain. The required range, 100Hz to 20kHz, is covered by the measured range. The measured efficiency at 1% THD is 55.4%, greater than the 40% threshold. The measured max gain is 26.2dB, greater than the 26dB threshold. The input and output impedances are 6053Ω and 0.08Ω respectively. Input impedance is greater than 1kΩ and output impedance is less than 1Ω. The audio amplifier is functional with supply voltages between 6V and 12V.

TABLE IV

PARAMETERS VS. MEASURED VALUES

Design Requirements	Measured Values
THD < 1%	0.21%
IRN < 75nV/√Hz	26.8nV/√Hz
PSRR > 60dB	65.6dB
CMRR > 60 dB	62.6dB
Gain Constant ± 3dB	Range: 89.9Hz – 449.9kHz
Efficiency > 40%	55.4%
Max Gain > 26 dB	26.2dB
AC Input Impedance > 1kΩ	6053Ω
AC Output Impedance < 1Ω	0.08Ω
Functionality?	Yes

REFERENCES

[1] B. Higgins, “Audio Amplifier Design and Simulation,” unpublished.
[2] “LM386 Low Voltage Audio Power Amplifier”, Texas Instruments Inc., Dallas, TX, USA, Rep. SNAS545C, May 2004.
[3] B. Higgins, “Basic Amplifier Simulations,” unpublished.
[4] B. Higgins, “Power Supply and Common Mode Rejection Ratio,” unpublished.
[5] Google LLC. *Gemini*. Accessed: Apr. 24, 2026. [Online]. Available: https://gemini.google.com/app
[6] B. Higgins, “Noise Simulations,” unpublished.
[7] B. Higgins, “Noise Overview,” unpublished.
[8] B. Higgins, “Advanced Simulations,” unpublished.
[9] B. Higgins, “ECE380 Amplifier PCB Design Package Instructions,” unpublished.